

A First-Triple Criterion for Log-Concavity of MDS Weight Distributions

Removing the dimension restriction by alternating-sum contraction

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Abstract

Let C be a linear $[n, k, d]_q$ maximum-distance-separable code, with $d = n - k + 1$, and let A_w count its words of Hamming weight w . Log-concavity is understood after zero coefficients have been deleted, with $A_0 = 1$ retained. We prove that, whenever $d \geq 2$, $k \geq 3$, and $q > d$, this entire distribution is log-concave if and only if $A_{d+1}^2 \geq A_d A_{d+2}$. There is no restriction on k/n or on n relative to q . The criterion is the single quadratic inequality

$$(n+1)q^2 - Bq + \frac{dB}{2} \geq 0, \quad B = k(d^2 + 2d - 1) + 2.$$

This removes the dimension restriction in the MDS result of Shi, Wang, An, and Kim and answers the high-dimension extension discussed in Remark 2 of their 2024 preprint. The proof normalizes the MDS weight formula into alternating negative-binomial partial sums. Their log-convexity defect contracts geometrically, fast enough to be absorbed by the binomial coefficient's log-concavity at every later weight. We also classify the boundary $q = d$, obtain an exact classification of single-parity-check codes, and derive sharp fixed-distance and fixed-rate thresholds. The accompanying standard-library Python programs reproduce exact checks on 154,888 main-case parameter triples, additional boundary cases, and 64 explicitly enumerated small codes. The results are proved for formal weight arrays as well as for existing codes; none is an assertion that arbitrary MDS parameters are realizable.

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1 Problem selection and the precise research target

1.1 The random draw

The area was selected before choosing a problem. A single call to Python’s `secrets.randbelow(24)` returned the zero-based index 16, selecting *coding theory*. The ordered list of candidate areas, the method, and the result are preserved in `data/random_selection.json`. No area was redrawn. This records the actual outcome, not a pseudorandom seed reconstructed after the choice.

1.2 The question from the literature

Shi, Wang, An, and Kim establish an MDS log-concavity criterion under $3 \leq k \leq n/2 + 3$; their Remark 2 discusses extending it to larger k [1, Theorem 9 and Remark 2]. The question addressed here is:

Does the same first-triple quadratic criterion suffice without the restriction $k \leq n/2 + 3$?

The answer proved below is affirmative for every integer $q > d$. The boundary $q = d$, where a weight coefficient vanishes, is treated separately rather than silently divided out.

The published version’s indexed abstract retains the stated dimension restriction [2]. Literature searches on 20 September 2026 found no subsequent removal of it. This is a bounded literature check, not a guarantee of priority. The full preprint was examined; the publisher’s full-text PDF could not be retrieved. Consequently the precise target and numbering above are anchored to the accessible preprint, not to an assumed identical journal layout.

1.3 What is, and is not, being solved

The main theorem supplies a full proof of this extension. It does not solve the MDS *existence* conjecture. Indeed, it does not use any conjectural upper bound on the length of an MDS code. The proof is an inequality argument for the universal MDS weight formula and is valid even for formal parameters for which no code exists.

Three distinctions matter throughout. First, we delete zero *coefficients*, not the weight-zero codeword. Second, a formal weight array is not a construction of a code. Third, an exact finite check corroborates the proof but cannot replace its universal argument. The discussion of missing weights by Ezerman, Grassl, and Solé provides useful background for these distinctions [3].

2 Definitions and the MDS weight formula

Definition 2.1. A linear $[n, k, d]_q$ code is a k -dimensional subspace of $\mathbb{F}_q^n$ whose smallest nonzero Hamming weight is d . It is *maximum-distance-separable*, or MDS, when $d = n - k + 1$. Write

$$A_w = \#\{c \in C : \text{wt}(c) = w\}, \quad W_C(z) = \sum_{w=0}^n A_w z^w.$$

The *compressed weight distribution* is the sequence of positive A_w , in increasing order of w , including $A_0 = 1$.

A positive sequence $(a_0, \dots, a_m)$ is log-concave if $a_i^2 \geq a_{i-1}a_{i+1}$ for every internal index. Equivalently, its successive quotients a_i/a_{i-1} are nonincreasing. The definition used here agrees with the coding-theoretic convention in [1]; it is not a statement about a sequence in which the missing weights remain as zeros.

For example, the binary even-weight code of length four has full array $(1, 0, 6, 0, 1)$ and compressed array $(1, 6, 1)$. These are different sequences for purposes of testing adjacent inequalities.

2.1 Derivation by shortening and inclusion–exclusion

An MDS generator matrix has every set of k columns independent. To see this, a dependency causing a rank drop in k selected columns would produce a nonzero codeword zero in those k positions, of weight at most $n - k = d - 1$, a contradiction. Every smaller set of columns is therefore independent as well.

Consequently, prescribing zeros in a set T of coordinates leaves $q^{k-|T|}$ words when $|T| \leq k$, and only the zero word when $|T| \geq k$. Fix a set S of $w \geq d$ positions. Inclusion–exclusion within S , after requiring zeros outside S , gives the number of words with support exactly S as

$$\sum_{j=0}^{w-d} (-1)^j \binom{w}{j} (q^{w-d+1-j} - 1). \quad (2.1)$$

The subtracted constants are legitimate because the complete alternating sum of the constant 1 vanishes when $w > 0$. Using Pascal's identity to telescope (2.1) gives

$$A_0 = 1, \quad A_w = 0 \ (0 < w < d), \quad A_w = (q-1) \binom{n}{w} \sum_{j=0}^{w-d} (-1)^j \binom{w-1}{j} q^{w-d-j} \quad (w \geq d). \quad (2.2)$$

This is the usual MDS weight formula, also the starting point of [1, Section 5]. Its derivation shows that all linear MDS codes with the same parameters have the same weight distribution.

2.2 Formal arrays and a basic feasibility constraint

Formula (2.2) defines an array for any integers $1 \leq k \leq n$ and $q \geq 2$, whether or not a code exists. We call this the *formal MDS array*. All statements about it below concern its numerical coefficients only.

The first three potentially nonzero weights are

$$A_d = (q-1) \binom{n}{d}, \quad (2.3)$$

$$A_{d+1} = (q-1) \binom{n}{d+1} (q-d), \quad (2.4)$$

$$A_{d+2} = (q-1) \binom{n}{d+2} \left(q^2 - (d+1)q + \frac{d(d+1)}{2} \right). \quad (2.5)$$

If $k \geq 2$, (2.4) exists in the array and implies the necessary feasibility condition $q \geq d$. If $q = d$, that coefficient is zero. If $q > d$, the proof below shows that *all* coefficients from A_d through A_n are positive.

3 Main theorem: one inequality decides everything

Theorem 3.1 (First-triple criterion). *Let n, k, q be integers, let $d = n - k + 1$, and suppose*

$$d \geq 2, \quad k \geq 3, \quad q \geq d + 1.$$

For the formal MDS array, the following are equivalent:

- (i) *The compressed distribution $(1, A_d, A_{d+1}, \dots, A_n)$ is log-concave.*
- (ii) *$A_{d+1}^2 \geq A_d A_{d+2}$.*
- (iii) *With $B = k(d^2 + 2d - 1) + 2$,*

$$P_{n,k}(q) := (n+1)q^2 - Bq + \frac{dB}{2} \geq 0. \quad (3.1)$$

- (iv) *$q \geq q_0(n, k)$, where*

$$q_0(n, k) = \frac{B + \sqrt{B(k-2)(d^2 - 1)}}{2(n+1)}. \quad (3.2)$$

When these conditions hold, every internal log-concavity inequality other than the one centered at A_{d+1} is strict. In particular, the theorem applies to every existing linear MDS code with these parameters, without a length or dimension restriction.

The proof occupies the next two sections. Its only analytical input is a simple bound on the ratios of successive terms in an alternating sum. No coding-theoretic classification theorem is used.

4 Alternating sums and normalized defects

4.1 A normalization that fixes the truncation parameter

Set $s = w - d$ and define

$$B_s = \sum_{j=0}^s (-1)^j \binom{d+s-1}{j} q^{s-j}. \quad (4.1)$$

Thus $A_{d+s} = (q-1) \binom{n}{d+s} B_s$. The symbol B_s denotes a sequence and is distinct from the coefficient B in (3.1). Pascal's identity gives

$$B_0 = 1, \quad B_s = (q-1)B_{s-1} + (-1)^s \binom{d+s-2}{s}. \quad (4.2)$$

For completeness, in (4.1) expand $\binom{d+s-1}{j} = \binom{d+s-2}{j} + \binom{d+s-2}{j-1}$. The first sum is qB_{s-1} plus its last term, and the second is $-B_{s-1}$.

Divide (4.2) by $(q-1)^s$. Define

$$t_j = \frac{\binom{d+j-2}{j}}{(q-1)^j}, \quad T_s = \sum_{j=0}^s (-1)^j t_j. \quad (4.3)$$

Here $t_0 = 1$, and we obtain the exact identities

$$B_s = (q-1)^s T_s, \quad A_{d+s} = (q-1)^{s+1} \binom{n}{d+s} T_s. \quad (4.4)$$

The crucial feature is that t_j no longer depends on the final truncation index s . It is a single negative-binomial sequence. The infinite alternating series sums to $((q-1)/q)^{d-1}$, but its value is not needed for the proof.

Lemma 4.1 (Term contraction and positivity). *Under the hypotheses of theorem 3.1,*

$$0 < t_1 < 1, \quad \frac{t_j}{t_{j-1}} = \frac{d+j-2}{j(q-1)} \leq \frac{1}{2} \quad (j \geq 2). \quad (4.5)$$

All T_s are positive. The odd partial sums are nondecreasing and satisfy $T_s \geq T_1$ for odd $s \geq 1$.

Proof. We have $t_1 = (d-1)/(q-1) < 1$. Since $q-1 \geq d$, the ratio in (4.5) is at most $(d+j-2)/(jd)$. The latter is at most $1/2$ because $(d-2)(j-2) \geq 0$. Successive pairs satisfy $t_{2r} - t_{2r+1} \geq 0$, so

$$T_{2r+1} = T_1 + \sum_{i=1}^r (t_{2i} - t_{2i+1}) \geq T_1 > 0.$$

Every even partial sum exceeds the next odd partial sum. This proves all the assertions and, by (4.4), the promised positivity of all A_w for $w \geq d$. $\square$

4.2 The competing effects

For an internal index $1 \leq s \leq k-2$, write $w = d+s$. The binomial coefficient contributes the factor

$$\frac{\binom{n}{w}^2}{\binom{n}{w-1}\binom{n}{w+1}} = \frac{(w+1)(n-w+1)}{w(n-w)} = 1 + E_s, \quad E_s := \frac{n+1}{(d+s)(n-d-s)}. \quad (4.6)$$

The powers of $q-1$ in (4.4) cancel in an adjacent ratio. Thus

$$\frac{A_{d+s}^2}{A_{d+s-1}A_{d+s+1}} = (1 + E_s) \frac{T_s^2}{T_{s-1}T_{s+1}}. \quad (4.7)$$

For even s , $T_s > T_{s-1}$ and $T_s > T_{s+1}$, so the second factor is greater than one. Every even-offset inequality is automatically strict.

For odd s , instead,

$$T_{s-1} = T_s + t_s, \quad T_{s+1} = T_s + t_{s+1}.$$

The alternating sum contributes a log-convexity defect

$$D_s := \frac{T_{s-1}T_{s+1}}{T_s^2} - 1 = \frac{t_s + t_{s+1}}{T_s} + \frac{t_s t_{s+1}}{T_s^2} > 0. \quad (4.8)$$

Consequently

$$\frac{A_{d+s}^2}{A_{d+s-1}A_{d+s+1}} = \frac{1 + E_s}{1 + D_s}, \quad A_{d+s}^2 \geq A_{d+s-1}A_{d+s+1} \iff D_s \leq E_s. \quad (4.9)$$

It is this comparison of *defect to curvature*, not monotonicity of the raw adjacent ratios, that works in every dimension.

Lemma 4.2 (Uniform defect comparison). *For every odd internal $s \geq 1$,*

$$\frac{D_s}{E_s} \leq \frac{s}{2^{s-1}} \frac{D_1}{E_1}. \quad (4.10)$$

Proof. Let $c = 2^{1-s}$. By theorem 4.1,

$$t_s \leq ct_1, \quad t_{s+1} \leq ct_2, \quad T_s \geq T_1.$$

Substituting into (4.8) gives

$$D_s \leq c \frac{t_1 + t_2}{T_1} + c^2 \frac{t_1 t_2}{T_1^2} \leq c D_1.$$

On the other hand, $d + s \leq s(d + 1)$ and $n - d - s \leq n - d - 1$, with all relevant factors positive. Hence

$$\frac{E_s}{E_1} = \frac{(d + 1)(n - d - 1)}{(d + s)(n - d - s)} \geq \frac{1}{s}.$$

Combining these two bounds proves (4.10). $\square$

For $s = 1$, the multiplier in (4.10) is one; for every odd $s \geq 3$ it is strictly smaller than one. In particular, the first odd defect is the only possible obstruction to *all* log-concavity inequalities holding simultaneously. Later inequalities can certainly fail when the first fails; the theorem does not say that a non-log-concave array has only one failure.

5 Proof of the criterion and the sharp threshold

5.1 The boundary involving the zero word

There is one internal compressed inequality not covered by (4.7): the one centered at A_d , whose left neighbor is $A_0 = 1$. By (2.3)–(2.4),

$$\frac{A_d^2}{A_{d+1}} = \frac{q-1}{q-d} \binom{n}{d} \frac{d+1}{n-d} > 1. \quad (5.1)$$

For example, $\binom{n}{d} \geq n$ for $1 \leq d \leq n-1$, and $(q-1)/(q-d) \geq 1$, so the strict inequality is immediate.

5.2 From the first triple to every triple

Condition (i) in theorem 3.1 plainly implies (ii). Conversely, (ii) is $D_1 \leq E_1$ by (4.9). For every later odd internal s , theorem 4.2 gives

$$\frac{D_s}{E_s} \leq \frac{s}{2^{s-1}} < 1.$$

All even-offset inequalities are strict by the discussion following (4.7), and the zero-word boundary is strict by (5.1). This proves (ii) $\Rightarrow$ (i), together with the theorem's assertion about equality.

5.3 Reduction to one quadratic

Substitute (2.3)–(2.5) into (ii) and cancel positive factors. Since $n - d = k - 1$, the result is

$$(d+2)(k-1)(q-d)^2 - (d+1)(k-2) \left(q^2 - (d+1)q + \frac{d(d+1)}{2} \right) \geq 0. \quad (5.2)$$

The coefficient of q^2 is $d + k = n + 1$; the coefficient of q is $-B$; and the constant is $dB/2$. Thus (ii) and (iii) are equivalent. Notice that dB is even: if d is odd, then $d^2 + 2d - 1$ is even. The exact test may nevertheless use $2P$ to avoid any division.

A convenient simplification is

$$B - 2d(n + 1) = (k - 2)(d^2 - 1). \quad (5.3)$$

The discriminant is therefore $B(k - 2)(d^2 - 1) > 0$, and

$$P_{n,k}(d) = -\frac{d}{2}(k - 2)(d^2 - 1) < 0. \quad (5.4)$$

The smaller root lies below d and the larger root above it. Since $q > d$, (iii) is equivalent to being at or above the larger root, which is exactly (3.2). This proves (iii) $\Leftrightarrow$ (iv) and completes the proof of theorem 3.1.

5.4 Why the dimension restriction disappears

The earlier argument compares adjacent log-concavity ratios along odd offsets [1, proof of Theorem 9]. Their binomial contribution need not increase before the middle of the weight range. In fact, when q tends to infinity with n, d fixed, those ratios approach the binomial ratios in (4.6), which decrease on part of the range. Thus a proof based on their universal monotonicity would be inappropriate.

The normalized quantity D_s/E_s asks a different question: how large is the harmful defect relative to the available positive curvature? The defect loses a factor at least 2^{s-1} , whereas the curvature can lose at most a factor s relative to the first position. The comparison $s \leq 2^{s-1}$ is independent of k/n . That is the entire mechanism behind the extension.

6 The boundary and a complete conditional classification

It is useful to remove the hypothesis $q > d$ carefully. The resulting classification is conditional on code existence, but the proof again works for formal arrays. In particular, statements about parameters with $q = d$ and large k below do not assert that such MDS codes exist.

Theorem 6.1 (Boundary cases). *Let $1 \leq k \leq n$, $d = n - k + 1$, and let $q \geq 2$ be an integer.*

- (i) *If $k = 1$ or $d = 1$, the compressed formal MDS array is log-concave.*
- (ii) *For $k \geq 2$, $q < d$ is impossible for an MDS code because the formal coefficient A_{d+1} is negative.*
- (iii) *If $k = 2$ and $q \geq d$, the compressed array is log-concave.*
- (iv) *If $q = d = 2$, the compressed array is log-concave for every n .*
- (v) *If $q = d \geq 3$ and $k \geq 3$, the compressed array is log-concave exactly when*

$$k \leq 4, \quad \text{or} \quad k = 5 \text{ and } d \geq 9.$$

Together with theorem 3.1, this decides compressed log-concavity for every existing nonzero linear MDS code.

Proof. If $k = 1$, the compressed array is $(1, q - 1)$. If $d = 1$, the code is the full space and $A_w = \binom{n}{w}(q - 1)^w$, a log-concave sequence. Part (ii) follows from (2.4). If $k = 2$, there are at most

two nonzero positive-weight coefficients. When $q > d$, their only internal inequality is (5.1); when $q = d$, only A_d remains.

For $d = q = 2$, (4.2) yields $B_s = 1$ for even s and $B_s = 0$ for odd s . Thus $A_w = \binom{n}{w}$ for even w and $A_w = 0$ for odd w . The compressed subsequence is log-concave: the quotient

$$\frac{\binom{n}{2r+2}}{\binom{n}{2r}} = \frac{(n-2r)(n-2r-1)}{(2r+2)(2r+1)}$$

is decreasing in r .

Now let $q = d \geq 3$. The normalized terms satisfy $T_1 = 0$, but $t_j/t_{j-1} < 1$ for $j \geq 2$. Hence $T_s > 0$ for all $s \geq 2$: $T_2 = t_2 > 0$, $T_3 = t_2 - t_3 > 0$, and the later alternating pairs preserve positivity. Therefore the only missing positive weight is $d+1$. The first support-count factors are

$$\begin{aligned} B_0 &= 1, & B_1 &= 0, & B_2 &= \frac{d(d-1)}{2}, \\ B_3 &= \frac{d(d-1)(d-2)}{3}, & B_4 &= \frac{d(d-1)(3d^2-7d+6)}{8}. \end{aligned} \quad (6.1)$$

We first check the two compressed inequalities affected by deleting A_{d+1} . When $k \geq 3$,

$$\frac{A_d^2}{A_{d+2}} = \frac{2\binom{n}{d}^2}{d\binom{n}{d+2}} > 1.$$

Indeed, $2 \leq d \leq n-2$ gives $\binom{n}{d} \geq \binom{n}{d+2}$, and therefore $\binom{n}{d}^2/\binom{n}{d+2} \geq (d+1)(d+2)/2$. When $k \geq 4$,

$$\begin{aligned} \frac{A_{d+2}^2}{A_d A_{d+3}} &= \frac{(k-1)(k-2)(d+3)}{(d+1)(d+2)(k-3)} \frac{3d(d-1)}{4(d-2)} \\ &\geq \frac{9}{2} \frac{d(d-1)(d+3)}{(d-2)(d+1)(d+2)} > 1. \end{aligned}$$

Here $(k-1)(k-2)/(k-3) \geq 6$, and the difference between the numerator and denominator in the last fraction is $d^2 + d + 4 > 0$. Thus $k = 3, 4$ are settled.

For $k \geq 5$, the next internal inequality is centered at A_{d+3} . Using (6.1), it becomes

$$U_d(k-3) - V_d(k-4) \geq 0, \quad (6.2)$$

where

$$U_d = 16(d-2)^2(d+4), \quad V_d = 9(3d^2-7d+6)(d+3).$$

We have

$$V_d - U_d = 11d^3 + 18d^2 + 57d - 94 > 0$$

for $d \geq 3$. The left side of (6.2) decreases with k . At $k = 6$ it equals

$$3U_d - 2V_d = -6d^3 - 36d^2 - 306d + 444 < 0.$$

Every $k \geq 6$ therefore fails this one inequality.

For $k = 5$, (6.2) is the last remaining internal inequality, and it is

$$f(d) := 5d^3 - 18d^2 - 249d + 350 \geq 0.$$

The values at $d = 3, 4, 5$ are negative, while $f(d+1) - f(d) = 15d^2 - 21d - 262 > 0$ for $d \geq 5$. Finally $f(8) = -234$ and $f(9) = 296$. Hence, for integer $d \geq 3$, this condition is exactly $d \geq 9$. The proof is complete. $\square$

Remark 6.2. The quadratic criterion is intentionally not applied at $q = d$. For example, the $[6, 3, 4]_4$ array is $A_0 = 1, A_4 = 45, A_6 = 18$, which is compressed-log-concave although $A_5^2 < A_4 A_6$. Testing the zero coefficient as though it had not been deleted would answer a different question.

7 Exact classification of single-parity-check codes

For $n \geq 2$, let

$$C_{n,q} = \{(x_1, \dots, x_n) \in \mathbb{F}_q^n : x_1 + \dots + x_n = 0\}.$$

This is an $[n, n-1, 2]_q$ MDS code for every n : its dimension is $n-1$, it has no word of weight one, and it contains a word of weight two. Thus there is no length-existence issue in this family.

A direct count gives

$$A_w = \binom{n}{w} \frac{(q-1)^w + (q-1)(-1)^w}{q}. \quad (7.1)$$

For a short derivation, let N_w count w -tuples of nonzero elements summing to zero. A tuple of length $w-1$ with nonzero sum has exactly one nonzero last coordinate that makes its total zero. Hence $N_w = (q-1)^{w-1} - N_{w-1}$ with $N_1 = 0$, which solves to the factor in (7.1).

Corollary 7.1. *The compressed weight distribution of $C_{n,q}$ is log-concave exactly in the following cases, where q is a prime power:*

Alphabet size	Permitted lengths $n \geq 2$
$q = 2$	Every length
$q = 3$	$n \leq 3$
$q = 4$	$n \leq 6$
$q = 5$	$n \leq 15$
$q \geq 7$	Every length

Proof. The binary case is theorem 6.1. For $q \geq 3$ and $n \geq 4$, substitute $d = 2, k = n-1$, and $B = 7n-5$ into (3.1):

$$P_{n,n-1}(q) = n(q^2 - 7q + 7) + (q^2 + 5q - 5). \quad (7.2)$$

At $q = 3, 4, 5$, this is respectively $19 - 5n, 31 - 5n$, and $45 - 3n$. For $q \geq 7$, both coefficients in (7.2) are positive. The omitted lengths $n = 2, 3$ have dimension at most two and are covered by theorem 6.1. $\square$

The threshold at $q = 5$ exhibits genuine equality. At length 15,

$$A_2 = 420, \quad A_3 = 5460, \quad A_4 = 70980, \quad A_5^2 = A_2 A_4.$$

Every other internal inequality is strict. At length 16,

$$A_2 = 480, \quad A_3 = 6720, \quad A_4 = 94640,$$

and $A_5^2 - A_2 A_4 = -268800$. The failure occurs near the beginning of the nonzero weight range, not near the peak of the distribution.

8 Further consequences and asymptotic regimes

8.1 An exact dimension cutoff at fixed distance

Fix $d \geq 2$ and an integer $q > d$, and put

$$h = d^2 + 2d - 1, \quad R_d(q) = q^2 - hq + \frac{dh}{2}, \quad S_d(q) = dq^2 - 2q + d. \quad (8.1)$$

Since $n + 1 = k + d$, the quadratic in (3.1) separates as

$$P_{n,k}(q) = kR_d(q) + S_d(q). \quad (8.2)$$

Also $S_d(q) > 0$ for the stated range.

Corollary 8.1. *For fixed $d \geq 2$ and $q > d$, the formal MDS arrays are log-concave for every dimension if $R_d(q) \geq 0$. If $R_d(q) < 0$, the exact criterion for dimensions $k \geq 3$ is*

$$k \leq \left\lfloor \frac{S_d(q)}{-R_d(q)} \right\rfloor.$$

In particular, at fixed d, q , log-concavity cannot fail at one dimension and then return at a larger dimension.

Proof. Apply theorem 3.1 to (8.2). The small dimensions are already log-concave. In fact P at $k = 2$ is $(d + 2)(q - d)^2 > 0$, so the cutoff is never below two. $\square$

8.2 A sharp uniform alphabet threshold

The larger root of R_d is

$$Q_\infty(d) = \frac{h + \sqrt{h(d^2 - 1)}}{2}. \quad (8.3)$$

As $R_d(d) = -d(d^2 - 1)/2 < 0$, the condition $R_d(q) \geq 0$ in our range is exactly $q \geq Q_\infty(d)$. This is a sharp uniform threshold for the *formal* arrays: below it, increasing k eventually causes failure; at or above it, every dimension passes. Existence restrictions may of course prevent particular failing formal arrays from being realized as codes.

For fixed d , the root $q_0(n, k)$ strictly increases with k and converges to $Q_\infty(d)$. To see strict increase, at the root for k we have $kR_d + S_d = 0$, so $R_d < 0$; increasing k makes the polynomial negative at that old root. The positive root must move right. Dividing (8.2) by k gives the limit.

d	$Q_\infty(d)$, approximately	Smallest prime power at or above it
2	5.7913	7
3	12.2915	13
4	20.7871	23
5	31.2829	32
6	43.7793	47
7	58.2764	59
8	74.7739	79
9	93.2719	97
10	113.7702	121

Expansion of (8.3) gives

$$Q_\infty(d) = d^2 + \frac{3}{2}d - \frac{5}{4} + \frac{1}{4d} - \frac{9}{16d^2} + O(d^{-3}). \quad (8.4)$$

This follows by applying the Taylor expansion of $\sqrt{1-x}$ to $\sqrt{h(h-2d)}$; it is not used to decide any finite case.

8.3 Fixed positive rate: the threshold is quadratic in length

Corollary 8.2. *Suppose $n \rightarrow \infty$ and $k/n \rightarrow \rho \in (0, 1)$. Then*

$$\frac{q_0(n, k)}{n^2} \rightarrow \rho(1 - \rho)^2. \quad (8.5)$$

In particular, a sequence of such MDS codes over alphabets of size $q = O(n)$ with $q > d$ eventually fails compressed log-concavity.

Proof. We have $d/n \rightarrow 1 - \rho$ and $B/n^3 \rightarrow \rho(1 - \rho)^2$. Rewrite (3.2), using (5.3), as

$$q_0(n, k) = \frac{B}{2(n+1)} \left(1 + \sqrt{1 - \frac{2d(n+1)}{B}} \right).$$

The fraction inside the square root tends to zero. Dividing by n^2 proves (8.5). A linear alphabet size is eventually smaller than this positive quadratic threshold. $\square$

This is an actual infinite-family consequence, not just a formal one. Take prime powers $q \rightarrow \infty$, set $n = q$, and evaluate all polynomials of degree less than $k = \lfloor \rho q \rfloor$ at every element of $\mathbb{F}_q$. A nonzero polynomial has at most $k - 1$ roots, and a product of $k - 1$ distinct linear factors attains that number. The resulting Reed–Solomon code has parameters $[q, k, q - k + 1]_q$. For large q , $q > d$, and theorem 8.2 proves its failure of compressed log-concavity.

8.4 Why a stronger normalized property does not follow

Normalize the positive-weight coefficients by their binomial factors: $a_s = A_{d+s} / \binom{n}{d+s}$. For $d \geq 2$, $k \geq 3$, $q > d$, (4.4) and (4.8) give

$$\frac{a_1^2}{a_0 a_2} = \frac{1}{1 + D_1} < 1.$$

Thus the binomial-normalized positive-weight sequence is *never* log-concave in this range, even when the ordinary weight counts are. The binomial factor is doing essential work.

Nor does ordinary log-concavity imply real-rootedness of the compressed generating polynomial. For the $[5, 3, 3]_7$ parameters, that polynomial is

$$1 + 60z + 120z^2 + 162z^3.$$

Its coefficients are log-concave, but its cubic discriminant is -74753388 , so it has a nonreal conjugate pair of roots.

9 Examples, computation, and reproducibility

9.1 Selected exact examples

The following table records values of the integer polynomial $2P$. For every row, $q > d$ and $k \geq 3$, so its sign is the full criterion.

n	k	d	q	$2P_{n,k}(q)$	Compressed log-concavity
5	3	3	5	-8	No
5	3	3	7	104	Yes
11	9	3	11	472	Yes
4	3	2	3	-2	No
7	6	2	4	-8	No
15	14	2	5	0	Yes; first-triple equality
16	15	2	5	-6	No

The $[11, 9, 3]_{11}$ example has $9 > 11/2 + 3$ and is directly constructed by polynomial evaluation over $\mathbb{F}_{11}$. Its first three positive-weight coefficients are

$$A_3 = 1650, \quad A_4 = 26400, \quad A_5 = 383460,$$

with $A_4^2 - A_3A_5 = 64251000 > 0$.

For the boundary example $[6, 3, 4]_4$, the archive explicitly enumerates the code generated over $\mathbb{F}_4 = \mathbb{F}_2[\omega]$, $\omega^2 = \omega + 1$, by

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & \omega & \omega^2 \\ 0 & 0 & 1 & 1 & \omega^2 & \omega \end{pmatrix}.$$

Its systematic form gives 64 distinct words. Their exact weight array is $(1, 0, 0, 0, 45, 0, 18)$.

9.2 An exact and inexpensive decision procedure

For the main range, no codeword enumeration and no floating-point root computation are necessary. Calculate

$$d = n - k + 1, \quad B = k(d^2 + 2d - 1) + 2,$$

then test the sign of

$$2(n+1)q^2 - 2Bq + dB. \tag{9.1}$$

This uses a constant number of arbitrary-precision integer operations. Its bit cost depends polynomially on the bit lengths of n, k, q ; the statement is not that multiplication of unbounded integers has constant cost. Generating the full array instead takes k recurrence steps in (4.2), besides computing the binomial coefficients.

```
def twice_quadratic(n: int, k: int, q: int) -> int:
    d = n - k + 1
    b = k * (d*d + 2*d - 1) + 2
    return 2*(n + 1)*q*q - 2*b*q + d*b
```

The complete implementation in `code/mds.py` validates inputs, handles all boundary cases, computes the full array, and reports any directly detected compressed log-concavity failures.

9.3 Executed verification, with scopes separated

All reported checks were executed using Python 3.13.5 and exact integers or `fractions.Fraction`. No numerical tolerance was used in a pass/fail decision.

Check	Number
Main-case formal parameter triples, $q > d$	154,888
Formal boundary triples, $q = d \geq 3$	15,642
Binary boundary lengths	499
Degenerate-case parameter checks	1,633
Independent inclusion–exclusion comparisons	1,815
Exact rational checks of (4.10)	45,066
Explicitly enumerated small codes	64
Codewords enumerated across those codes	20,798

The main sweep uses every $4 \leq n \leq 200$ and $2 \leq d \leq n - 2$, with $k = n - d + 1$. For each pair it tests the distinct values in

$$\{d + 1, d + 2, 2d, 3d, n - 1, n, n^2, d^2 + 2d - 1\}$$

that exceed d . For every triple it compares the full compressed array against the theorem, checks positivity on the entire claimed support, checks that the total is q^k , and verifies that any failure includes the first triple. These triples include non-prime-power q and unrealizable code parameters deliberately: the theorem is stronger than a statement restricted to known constructions.

The $q = d \geq 3$ sweep uses $3 \leq d \leq 200$ and $2 \leq k \leq 80$. The binary checks cover lengths 2 through 500. The rational checks test the central bound directly with exact fractions for $n \leq 60$.

The 64 construction checks are separate. They enumerate small prime-field Reed–Solomon codes, single-parity-check codes over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_5, \mathbb{F}_7$, and the displayed $\mathbb{F}_4$ code. Their word counts are compared with the universal formula. The `verification.json` file records every constructed code’s parameters and the full selected-example arrays.

9.4 Reproducing the archive

The source requires an ordinary L^AT_EX installation with the standard packages listed in the preamble. From the archive root:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python code/verify.py --max-n 200 --output data/verification.json
python code/mds.py 11 9 11
```

Alternatively run `pdflatex` repeatedly until references stabilize. The Python programs require version 3.9 or later and only the standard library. The verification runtime was approximately 21 seconds in the authoring environment; it naturally varies between machines. The deterministic digest of the main sweep is included in the JSON output, so the substantive record can be compared independently of runtime or Python-version metadata.

10 Conclusion and status

The selected extension has an affirmative answer: the first-triple quadratic criterion does not require a dimension restriction. The key step is (4.10), which compares a geometric alternating-sum defect with a binomial curvature that can deteriorate only linearly relative to the first position.

The treatment also closes the zero-coefficient boundary algebraically, yields an exact classification for every existing linear MDS code, and gives explicit length thresholds for single-parity-check codes. At fixed distance there is a finite uniform alphabet threshold; at fixed positive rate the necessary alphabet threshold grows quadratically with length. These different regimes explain why apparently similar families can have very different log-concavity behavior.

The article supplies a conventional proof and executed exact-arithmetic checks. It has not been proof-assistant formalized or independently peer reviewed. No gap is known in the argument presented here, but novelty and correctness remain subject to independent scrutiny. The priority claim is limited to the literature check described in Section 1; the explicit theorem and proof do not depend on that claim.

A A compact proof for independent checking

For a reader auditing the central result without the surrounding applications, the proof can be compressed to the following chain. Let $d \geq 2$, $k \geq 3$, $q \geq d + 1$ and $n = d + k - 1$. Define

$$t_j = \frac{\binom{d+j-2}{j}}{(q-1)^j}, \quad T_s = \sum_{j=0}^s (-1)^j t_j.$$

The MDS recurrence proves $A_{d+s} = \binom{n}{d+s} (q-1)^{s+1} T_s$. Because $t_1 < 1$ and $t_j/t_{j-1} \leq 1/2$ for $j \geq 2$, all T_s are positive and all odd $T_s \geq T_1$.

Even offsets give strict log-concavity immediately. For odd s , set

$$D_s = \frac{t_s + t_{s+1}}{T_s} + \frac{t_s t_{s+1}}{T_s^2}, \quad E_s = \frac{n+1}{(d+s)(n-d-s)}.$$

The adjacent inequality is exactly $D_s \leq E_s$. Term contraction gives $D_s \leq 2^{1-s} D_1$; elementary denominator comparison gives $E_s \geq E_1/s$. Therefore

$$D_1 \leq E_1 \implies \frac{D_s}{E_s} \leq \frac{s}{2^{s-1}} \leq 1.$$

The zero-word boundary is strict because $A_d^2/A_{d+1} > 1$. Finally, substituting the first three weights reduces $D_1 \leq E_1$ to (3.1); (5.4) identifies the relevant root without any MDS existence bound.

B Artifact inventory and provenance

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	This article and its editable source.
<code>code/mds.py</code>	Exact enumerator, full criterion, command-line examples.
<code>code/verify.py</code>	Reproducible formal and actual-code checks.
<code>code/make_tables.py</code>	Exact threshold selection and table regeneration.
<code>data/verification.json</code>	Executed results, scope, examples, deterministic digest.
<code>data/random_selection.json</code>	The area-selection list, method, and draw.
<code>data/thresholds.csv</code>	Numerical values of the exact uniform threshold.
<code>data/threshold_rows.tex</code>	Generated rows mirrored in the article source.
<code>data/literature_audit.json</code>	Sources, exact target, search date, access limitations.
<code>README.md</code>	Build and verification instructions.

The external source articles are not bundled. Their references are listed below, and the numerical and mathematical work in this archive can be inspected without downloading them.

References

- [1] Minjia Shi, Xuan Wang, Junmin An, and Jon-Lark Kim. *Log-Concave Sequences in Coding Theory*. arXiv:2410.04412v1, 6 October 2024, 31 pages. Especially Section 5, Theorem 9, and Remark 2 on pages 27–29. <https://arxiv.org/abs/2410.04412>.
- [2] Minjia Shi, Xuan Wang, Junmin An, and Jon-Lark Kim. *Log-Concave Sequences in Coding Theory*. IEEE Transactions on Information Theory, 71(12):9516–9533, 2025. Publisher record 11176928. Bibliographic record and indexed abstract consulted; full journal PDF not retrieved. <https://ieeexplore.ieee.org/document/11176928/>.
- [3] Martianus Frederic Ezerman, Markus Grassl, and Patrick Solé. *The Weights in MDS Codes*. IEEE Transactions on Information Theory, 57(1):392–396, 2011. Preprint arXiv:0908.1669. <https://arxiv.org/abs/0908.1669>.